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QMLSC: A quantum multimodal learning model for sentiment classification

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ABSTRACT

Sentiment classification research is gaining prominence for enhancing user experience, facilitating targeted marketing, and supporting mental health assessments while driving technological innovation. Due to the complexity and diversity of emotional expression, this study proposes quantum multimodal learning for sentiment classification (QMLSC), a novel quantum–classical hybrid model that integrates text and speech data to capture emotional signals more effectively. To address the limitations of the noisy intermediate-scale quantum era, we designed advanced variational quantum circuit (VQC) architectures to efficiently process high-dimensional data, maximizing feature retention and minimizing information loss. Our approach employs a residual structure that fuses quantum and classical components, enhancing the benefits of quantum features and conventional machine learning attributes. By using randomized expressive circuits, we improve system flexibility, accuracy, and robustness in sentiment classification tasks. Integrating VQC significantly reduces the number of parameters compared to fully connected layers, resulting in improved accuracy and computational efficiency. Empirical findings validate the superior performance of our fusion approach in effectively mitigating noise and error impacts associated with quantum computing and demonstrate strong potential for future applications in complex emotional information processing. This study provides new insights and methodologies for advancing sentiment classification technology and highlights the broad application potential for advancing quantum computing in information processing fields.

1. Introduction

Sentiment classification is crucial in domains such as enhancing user experience, targeted marketing and advertising, mental healthcare, and educational technology [1]. Traditional sentiment analysis models primarily focus on textual data, extracting sentiment by examining semantic relationships within sentences. However, with advancements in technology, individuals now express their opinions and emotions through diverse modalities, including audio, images, and video. This shift has highlighted the limitations of single-modal approaches, propelling multimodal fusion strategies to the forefront of research [2]. By transitioning from single-modality frameworks to multimodal paradigms, sentiment analysis has embraced new opportunities to integrate complementary data streams, enabling more robust sentiment detection and providing deeper insights. For example, subtle and complex expressions such as irony and exaggeration, often challenging to interpret through text alone, can be more effectively captured through multimodal analysis, demonstrating the significant potential of this research direction.

Quantum computing, a rapidly advancing field in computational science, demonstrates the substantial potential for addressing complex problems [3,4]. By leveraging quantum bits (qubits) based on principles like superposition and entanglement [5], quantum computing enables highly parallel computations, significantly enhancing efficiency and capacity compared to classical systems. This enhanced capability has driven revolutionary advancements in information processing, enabling breakthroughs in multimodal learning tasks such as sarcasm detection [6], fake news identification [7], and intelligent diagnosis [8]. Among these achievements, quantum neural networks (QNNs) have emerged as a prominent model at the intersection of quantum computing and classical machine learning, demonstrating significant potential for advanced applications. Compared to classical neural networks, QNNs exhibit an enhanced ability to model intricate and sophisticated feature representations [4,9]. Notably, QNNs show superior performance in capturing high-dimensional data characteristics, high-frequency components, and complex feature structures [10,11]. These strengths suggest that QNNs could also excel in sentiment classification tasks, where the ability to process subtle and complex data patterns is crucial.

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1566-2535/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Despite these, research on multimodal sentiment classification tasks leveraging quantum computing remains in its infancy. To address this gap, this paper proposes a quantum–classical hybrid multimodal sentiment classification model that combines text and speech data to enable a more comprehensive and nuanced analysis of emotional signals. By combining the linguistic depth of textual data with acoustic features such as pitch, speech rate, and volume derived from audio, this approach effectively bridges the gap between distinct yet complementary modalities. Furthermore, by incorporating quantum neural network, this approach can deeply explore and integrate the advantages of distinct emotional representations across different modalities, thereby significantly enhancing the accuracy and depth of sentiment recognition. Currently, quantum computers operate in the Noisy intermediate-scale quantum (NISQ) era [12,13], constrained by a limited number of qubits. This constraint often necessitates a significant reduction of high-dimensional data when employing angle encoding within QNNs, resulting in substantial feature loss. To tackle this issue, this study proposes a quantum data encoding circuit and a variational quantum circuit (VQC) search strategy, both specifically designed for the multimodal sentiment classification dataset. Furthermore, a residual fusion structure integrating quantum and classical neural networks is employed to enhance the performance of multimodal sentiment analysis [14].

The primary contributions of this study are outlined as follows.

- This study employs widely recognized VQC layers to optimize parameter size and employs random search to construct the VQC, enhancing feature identification and sentiment classification performance.
- A methodology is developed to incorporate additional features into quantum circuits despite the constraints of limited quantum resources.
- This research introduces a residual feature fusion structure that integrates quantum circuits with classical models, effectively mitigating the impact of quantum noise.
- We incorporated VQCs to significantly reduce the parameter count in fully connected layers, improving model accuracy.

The remainder of this paper is organized as follows. Section 2 provides the foundational concepts of quantum computing essential for understanding the work presented in this study. Section 3 reviews existing contributions in the fields of sentiment classification and quantum machine learning. The implementation framework for QMLSC is detailed in Section 4. To validate the feasibility and validity, Section 5 conducts an extensive experimental evaluation, followed by a detailed analysis and discussion. Finally, Section 6 concludes the paper and suggests directions for future research.

2. Fundamentals of quantum computing

2.1. Quantum bits (Qubits)

A qubit serves as the fundamental unit of quantum computing, analogous to a classical bit but possesses more complex and versatile properties. The state of a qubit can be mathematically described using Dirac notation as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{1}$$

where $\alpha, \beta \in \mathbb{C}$ satisfy $|\alpha|^2 + |\beta|^2 = 1$, and $|\cdot|$ denotes the modulus of a complex number. $|\alpha|^2$ and $|\beta|^2$ respectively represent the probabilities of the qubit being in the $|0\rangle$ and $|1\rangle$ states. Additionally, a qubit can also be represented on the Bloch sphere shown in Fig. 1 by rewriting Eq. (1) as

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\cos\frac{\theta}{2}|1\rangle. \tag{2}$$

Classical bits can only be in the state of 0 or 1, while quantum bits can be in a superposition of both 0 and 1 simultaneously, existing in multiple states at once. This characteristic enables quantum computers to process information in parallel, thereby demonstrating higher efficiency than classical computers in specific problems.

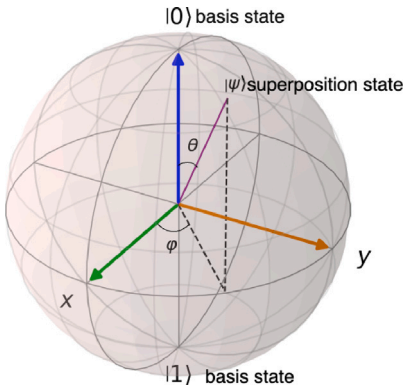


Fig. 1. The Bloch sphere representation of a single qubit.

2.2. Composite quantum systems and quantum entanglement

A composite quantum system is an entity formed by multiple independent quantum subsystems. Each subsystem possesses its own Hilbert space, and the composite system resides in the tensor product space of these individual Hilbert spaces. To illustrate this concept more clearly, we can consider a bipartite system composed of two quantum systems, A and B. Suppose the state vector of quantum system A, $|\psi_A\rangle$ belongs to the Hilbert space $\mathcal{H}_A$, and the state vector of quantum system B, $|\psi_B\rangle$ belongs to another Hilbert space $\mathcal{H}_B$. The combined state of these two systems can be represented by the tensor product of their respective states, denoted as

$$|\psi_{AB}\rangle = |\psi_A\,\psi_B\rangle = |\psi_A\rangle \otimes |\psi_B\rangle. \tag{3}$$

Here, the symbol $\otimes$ denotes the tensor product (also known as the Kronecker product), which generates a new Hilbert space $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$. However, when the superposition state of quantum bits cannot be expressed as a tensor product of individual qubit states, such a state is referred to as quantum entanglement. A specific example is one of the Bell states, which represents a maximally entangled state of two qubits

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle). \tag{4}$$

This special state cannot be decomposed into the tensor product of the states of two independent qubits, which is precisely the defining characteristic of an entangled state.

Quantum entanglement can be utilized to generate highly complex quantum states [15], which serve as inputs or feature spaces for machine learning models. By encoding classical data into entangled states, richer and more intricate data structures can be captured, thereby enhancing the performance of tasks such as classification and clustering. Although quantum entanglement offers substantial potential for quantum machine learning, practical applications still encounter significant challenges, including maintaining the stability of entangled states, mitigating noise effects, and addressing the limitations of current quantum hardware [13].

2.3. Quantum measurement

Quantum measurement is a fundamental process in quantum computing, involving the observation of a quantum system and leading to the collapse of its quantum state. For a quantum system described by the state $|\psi\rangle$ the measurement can be characterized by a set of measurement operators $\{M_m\}$, which act on the state space of the

system. The probability of obtaining a particular measurement outcome m is given by

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle, \tag{5}$$

where $M_m^\dagger$ is the conjugate transpose (i.e., Hermitian adjoint) of the measurement operator M_m, while $\langle\psi|$ and $|\psi\rangle$ are the bra and ket vectors, respectively, representing the inner product of the quantum state. The system state after measurement is

$$\frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}. \tag{6}$$

The measurement operators must satisfy the equations

$$\sum_m M_m^\dagger M_m = I, \ \sum_m p(m) = \sum_m \langle\psi|M_m^\dagger M_m|\psi\rangle = 1. \tag{7}$$

In quantum machine learning, classical data is typically encoded into quantum states. After processing through a series of quantum gate operations, the output results are decoded via measurement. Measurement serves as a bridge from quantum information back to classical information [16]. In variational quantum circuits, quantum measurements are used to evaluate the loss function, which is a critical step in optimizing model parameters. The measurement outcomes guide the adjustment of quantum gate angles or other parameters to minimize the loss function [17].

3. Related works

3.1. Sentiment classification

Early sentiment classification tasks predominantly employed dictionary-based and conventional machine learning approaches, which required large amounts of annotated data, making the process costly and labor-intensive. With advancements in neural networks, particularly deep learning technologies, researchers have shifted towards more effective methods that leverage semantic and syntactic information.

Semantic-based neural network approaches, such as long short-term memory (LSTM) networks and attention mechanisms, are extensively employed in sentiment classification tasks to capture the dependencies between focal words and their context. For instance, Tang et al. (2016) [18] introduced an LSTM-based method, and Wang et al. (2016) [19] incorporated LSTM with an attention mechanism to enhance focal word comprehension. Furthermore, the emergence of transformer models has further advanced sentiment analysis through multi-head attention mechanisms, refining the extraction of semantic features [20].

In contrast, grammar-based neural network approaches focus on dependency relationships within sentence structures by constructing graph representations to articulate word relationships. This methodology effectively reduces the distance between focus words and opinion words, improving the representational power of focal words. For instance, attention-based syntactic graph convolutional network [21] utilizes a grammar dependency matrix with a graph convolutional network (GCN) [22] to refine word representation. Additionally, robust graph models such as the graph attention network (GAT) [23] have been applied to sentiment classification, including the target-dependent graph attention network proposed by Ma et al. (2017) [24].

Researchers have integrated semantic and syntactic information to address the limitations of single-method approaches in sentiment classification. For instance, Gu et al. (2023) [25] and Lu et al. (2022) [26] employed heterogeneous graph neural networks to process sentence nodes and incorporate commonsense knowledge, enhancing text comprehension. Furthermore, to better capture interactions among focus words within a sentence, specialized models have been proposed. Majumder et al. (2018) [27] and Hazarika et al. (2018) [28] utilized memory networks and attention mechanisms, respectively, to elucidate sequential relationships among focus words. Gu et al. (2023) [25] introduced the IA-HiNET model, which leverages GCN and self-attention mechanisms to extract emotional information from focus words and model their interrelationships.

Current research in sentiment classification is advancing by combining semantic and syntactic information, using graph neural networks and pre-trained language models to refine the understanding of relationships among focus words, enhancing the accuracy and robustness of sentiment classification tasks. Elevating the precision of sentiment classification necessitates amalgamating text, audio, video, and other multi-dimensional data for a more comprehensive sentiment analysis. This approach has amplified the demand for multimodal sentiment detection, emphasizing the need for simultaneous and efficient analysis of diverse media content to capture complex emotional information embedded within.

3.2. Quantum machine learning

The concept of QNNs was first introduced by Kak et al. [29] in 1995, who pioneered the idea of quantum neural computing. Subsequently, Ventura et al. [30] proposed representing neural networks using quantum superposition states, highlighting that quantum computing can leverage microscopic quantum effects to execute computations significantly faster than classical methods in certain scenarios. Matsui et al. (2000) [31] explored quantum dynamics to characterize neural states and developed quantum bit neural networks, while Zhou et al. (2006) [32] introduced the quantum perceptron, utilizing quantum phase for computations, offering capabilities beyond those of traditional perceptrons. These early contributions laid the groundwork for the advancement of QNNs.

Recent advancements in quantum hardware have positioned variational quantum algorithms (VQAs) as a key research area, particularly QNNs based on quantum circuits. Farhi et al. (2018) [33] introduced a QNN tailored for binary classification with optimized parameters. Cong et al. (2019) [34] proposed a quantum convolutional neural network (QCNN) for classical image data processing. Du et al. (2021) [35] demonstrated the robustness of ANNs under depolarizing quantum noise during classification tasks. Qu et al. (2022) [36] designed a spatiotemporal quantum graph convolutional neural network for traffic congestion prediction, revealing resilience against interference. Qu et al. (2023) further highlighted the potential of QNNs in multimodal applications, including fake news detection and intelligent diagnosis [7, 8].

Quantum sentiment classification has emerged as a burgeoning research field. Zhang et al. (2020) [37] demonstrated the application of quantum probability theory in classification tasks. Meichanetzidis et al. (2023) [38] utilized the categorical distributional compositional (DisCoCat) model for syntactic classification, representing sentences and words as parameterized quantum circuits, with quantum entanglement depicting syntactic structures. Lorenz et al. (2021) [39] investigated mapping sentence representations onto quantum circuits within quantum environments, examining models such as “bag-of-words” that maintain syntactic independence. Gkoumas et al. (2021) [40] introduced a quantum cognitive model for predicting sentiment, expressing utterances as superpositions of positive and negative sentiments with univariate classifiers as mutually incompatible observables. Furthermore, Gkoumas et al. (2021) [41] and Liu et al. (2021) [42] proposed a transparent quantum probabilistic neural model and a multi-task learning framework driven by quantum probability.

Further research initiatives address challenges in sentiment classification using quantum technology. Some studies employ quantum-inspired models for sentiment analysis on Twitter data [43], evolving into quantum-like multimodal networks (QMNs). These networks integrate quantum theory with LSTM architectures for multimodal sentiment analysis in conversational contexts [37]. These advancements underscore the potential of quantum technology in natural language processing and the gradual emergence of viable solutions in quantum sentiment classification.

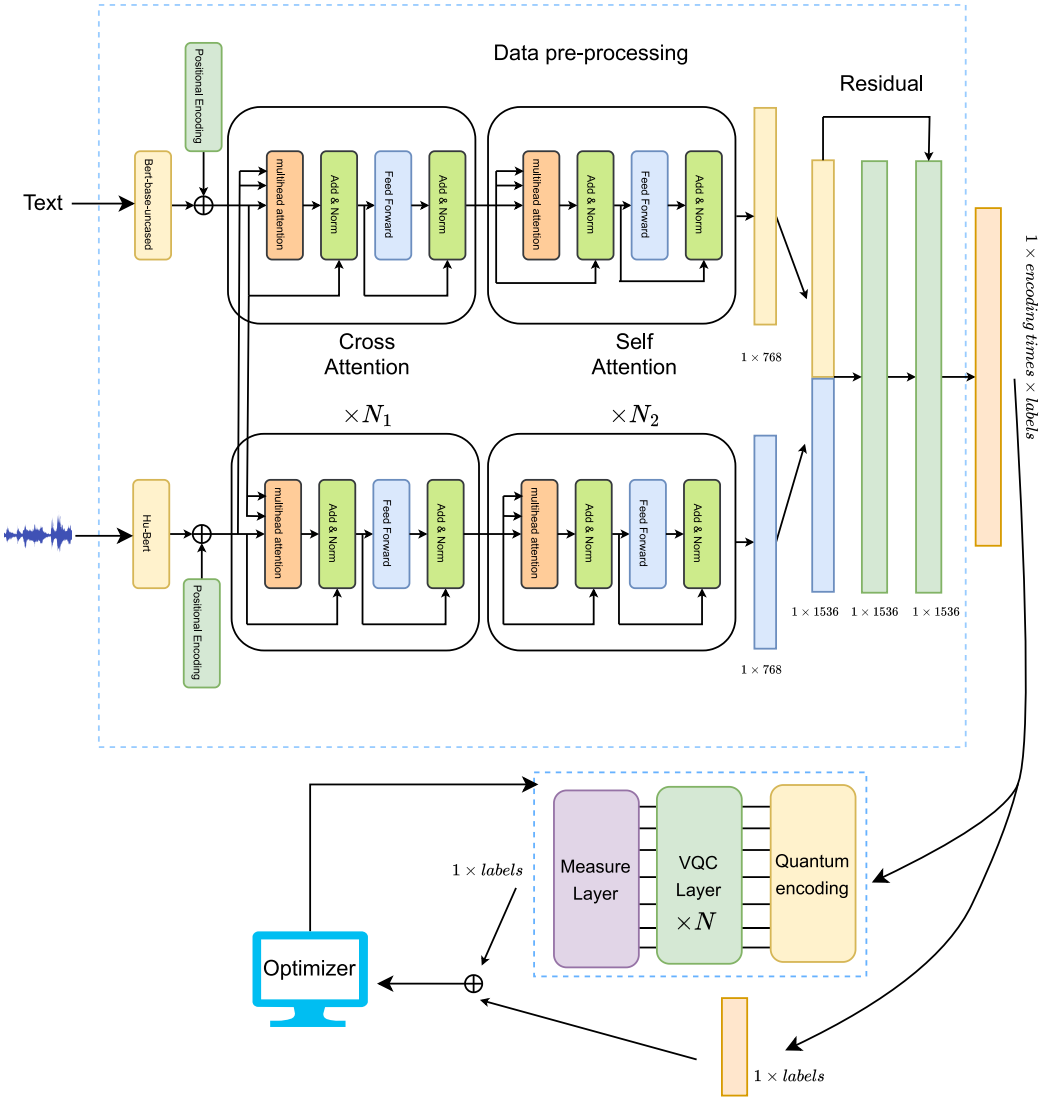


Fig. 2. The overall flowchart illustrating the QMLSC model.

4. Methodology

The VQC model operates in three stages, as depicted in the quantum multimodal learning model for sentiment classification (QMLSC) in Fig. 2. The first stage encodes original speech and text data into feature representations. In the second stage, these features are initially fused from both modalities into a classical neural network. The third stage encompasses further feature fusion through a combination of data encoding and quantum variational circuits. During this process, features are input into the quantum variational circuit, and measurement results are integrated with outputs from a classical fully connected layer to generate the final prediction. Model optimization occurs in the classical computing environment, where the loss function is computed, and parameters are adjusted accordingly.

4.1. Multimodal feature extraction

4.1.1. Text feature extraction

Text features play a crucial role in the model to grasp the nuanced meanings of textual data, including word semantics, contextual relationships, phrase structures, and sentence composition. Individual words may convey positive or negative emotional connotations, which are critical for accurately determining the overall sentiment orientation of the text.

This study employs BERT-base-uncased, a model developed by Google in 2018 for feature extraction [44]. BERT, a deep learning architecture based on Transformers, generates high-quality language representations through bidirectional context understanding. BERT comprises 12 Transformer encoder layers, each with 768 hidden layers and 12 attention heads. Pre-trained on an extensive and diverse text dataset, the BERT-base-uncased model captures general linguistic patterns through unsupervised learning, providing a robust basis for effective text feature extraction.

4.1.2. Audio feature extraction

Speech acoustic characteristics, such as pitch, loudness, speech rate, pauses, and duration, along with non-linguistic features, such as laughter and crying, are essential for sentiment classification.

This study employs a hierarchical hidden-unit back-projection transformer (HuBERT), a self-supervised learning model developed by Facebook AI Research (FAIR), to extract features for voice recognition and processing [45]. HuBERT learns audio representations across multiple abstract levels, capturing information at various temporal scales. The model is trained to predict masked audio segments, similar to the masked language modeling (MLM) task in BERT, but adapted for audio signals. HuBERT's multi-level feature extraction capabilities make it highly effective in speech-related tasks.

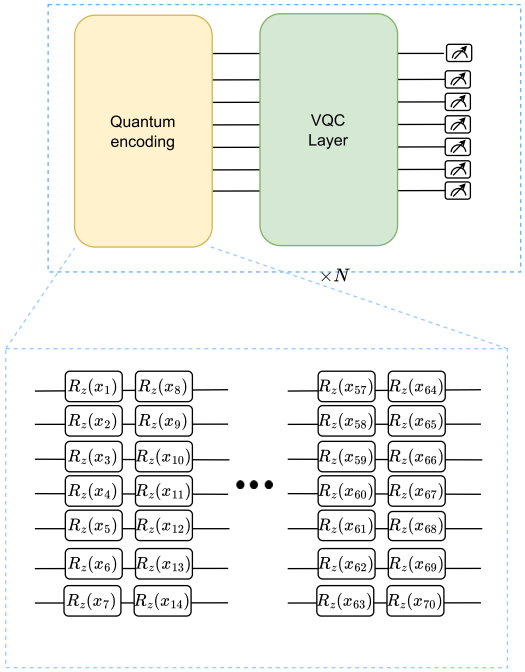


Fig. 3. Structure of the quantum encoding layer in QMLSC, demonstrated using an input of size 10 × 7.

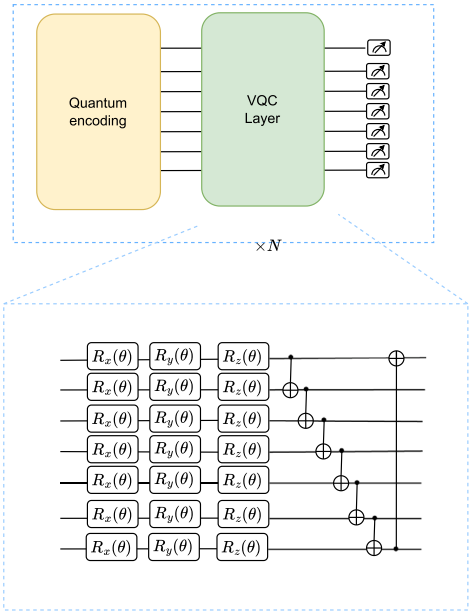


Fig. 4. Structure of the strong entanglement layer.

4.2. Feature pre-processing

This study utilizes the transformer architecture for processing text and speech sequence features, known for its high performance in sequence-related tasks. Both text and speech features are encoded into 768-dimensional representations. Initially, these features are processed through a 5-layer transformer with a cross-attention mechanism for feature extraction. Subsequently, these features are passed into a 3-layer transformer using a self-attention mechanism for further feature extraction. Finally, the output vectors at the positions corresponding to the [CLS] token are extracted from each of these two Transformer models as feature representations. The resulting 768-dimensional features are concatenated and fed into a fully connected layer.

4.3. Quantum–classical feature fusion

4.3.1. Quantum encoding layer

Unlike classical computers, quantum computers process data as quantum states, requiring classical data to be encoded into a quantum-compatible format. Quantum state encoding is widely used, with angle encoding being particularly effective in preserving the original feature distribution. Given a data vector $x = (x_1, x_2, x_3, \ldots, x_n)$, it can be transformed into a quantum state using various rotation gates such as: $R_x = \begin{pmatrix} \cos\theta/2 & -i\sin\theta/2 \\ -i\sin\theta/2 & \cos\theta/2 \end{pmatrix}$, $R_y = \begin{pmatrix} \cos\theta/2 & -\sin\theta/2 \\ -\sin\theta/2 & \cos\theta/2 \end{pmatrix}$, and $R_z = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$. For instance, R_x can be employed as the encoding gate, defined as

$$|x\rangle = \bigotimes_{i=1}^{n} R_x(x_i)|0\rangle. \tag{8}$$

However, current quantum computers are constrained by the limited availability of qubits, which restricts the number of features that can be utilized. Angle encoding applies rotation gates to each quantum bit; however, due to these constraints, only a small number of features can be processed within the quantum circuit. Inspired by quantum data re-uploading [46] and single qubit encoding [47], this study aims to enhance quantum circuit performance by incorporating additional features without increasing the original number of qubits. Features are encoded into corresponding quantum states via angle encoding and sequentially input into the quantum circuit in batches, facilitating the introduction of more effective features (Fig. 3). For example, given a vector $x = (0.1, 0.2, 0.3, 0.4)$ of size 1×4, encoding it into a quantum circuit consisting of 2 qubits produces a quantum state represented as

$$|x\rangle = (R_z(0.3) \otimes R_z(0.4))\,(R_z(0.1) \otimes R_z(0.2))\,(|0\rangle \otimes |0\rangle). \tag{9}$$

4.3.2. Variational quantum circuit layer

The design of VQCs is pivotal in defining the network's expressive capacity. Before establishing the search space, it is essential to estimate the network's parameter count. This study employs the strong entanglement layer shown in Fig. 4 [48], which is iteratively stacked to identify the requisite number of parameters for the quantum circuit based on accuracy as an evaluation metric. This optimal parameter count guides subsequent searches for circuit structures. Initially, the random search circuit determines the number of parameterized and non-parameterized quantum gates needed for exploration. Subsequently, it constructs circuits using randomized configurations to identify optimal structures. This process is detailed in Algorithm 1. For simplicity, the search space is constrained to a random layer of 21 parameterized gates and eight non-parameterized gates, repeated across 10 iterations to refine the search space further. The circuit structure optimized using Algorithm 1 on the MELD dataset [49] is illustrated in Fig. 5.

4.3.3. Quantum–classical residual fusion structure

The preprocessed features are partitioned into two streams. One stream is directed into the VQC, where quantum features are extracted using trainable quantum gates and specific measurement operators. The other stream is fed into a fully connected neural network layer for learning classical feature representations. As illustrated in Fig. 6, the entire structure can be described as

$$Output = Measure(U(\theta)E(x)|\psi_{initial}\rangle) + FC(x). \tag{10}$$

where $FC(x)$ is a fully connected neural network, $E(x)$ is the quantum encoding layer mapping the input x into a quantum state, and $U(\theta)$ is

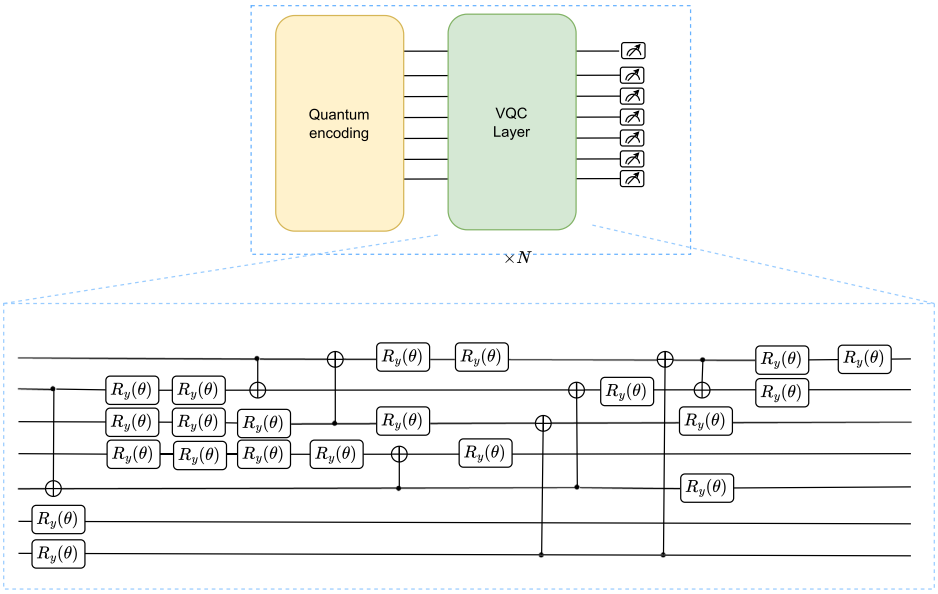


Fig. 5. The structure of the variational quantum circuit layer in QMLSC (exemplified by the model implemented on the MELD dataset).

Algorithm 1 Quantum Circuit Structure Optimization

Require:
ParamGates: Set of parameterized quantum gates
NonParamGates: Set of non-parameterized quantum gates
numParamGates: Number of parameterized gates to include
numNonParamGates: Number of non-parameterized gates to include
circuitDepth: Number of iterations for the quantum circuit
searchIterations: Number of iterations for the search algorithm
Ensure:
optimalCircuit: Structure of the optimal quantum circuit
Initialize s to store the optimal circuit structure determined
Initialize *temps* to temporarily store a circuit structure
Initialize *accuracy* to 0
for $i = 1$ **to** *searchIterations* **do**
  Reset *temps* to an empty circuit structure
  for $j = 1$ **to** *numParamGates* **do**
    $p1 \leftarrow$ Select a random gate from *ParamGates*
    $tp1 \leftarrow$ Select a random position in the quantum circuit for $p1$
    Add $p1$ at position $tp1$ to *temps*
  end for
  for $k = 1$ **to** *numNonParamGates* **do**
    $p2 \leftarrow$ Select a random gate from *NonParamGates*
    $tp2 \leftarrow$ Select a random position in the quantum circuit for $p2$
    Add $p2$ at position $tp2$ to *temps*
  end for
  taccuracy $\leftarrow$ Train a model using *temps* and *circuitDepth* to obtain accuracy
  if *taccuracy* > *accuracy* **then**
    $s \leftarrow$ *temps*
    accuracy $\leftarrow$ *taccuracy*
  end if
end for
return s

the variational quantum circuit parameterized by θ. The output features from both the quantum circuit and the fully connected neural network are subsequently fused, combining the strengths of both quantum and classical features. These fused features are processed through a softmax classifier to ascertain the probability distribution across different sentiment categories, selecting the category with the highest probability as the predicted outcome. The entire model is optimized using a cross-entropy loss function to minimize discrepancies between predicted and true emotion labels, enhancing accuracy and efficiency in sentiment classification.

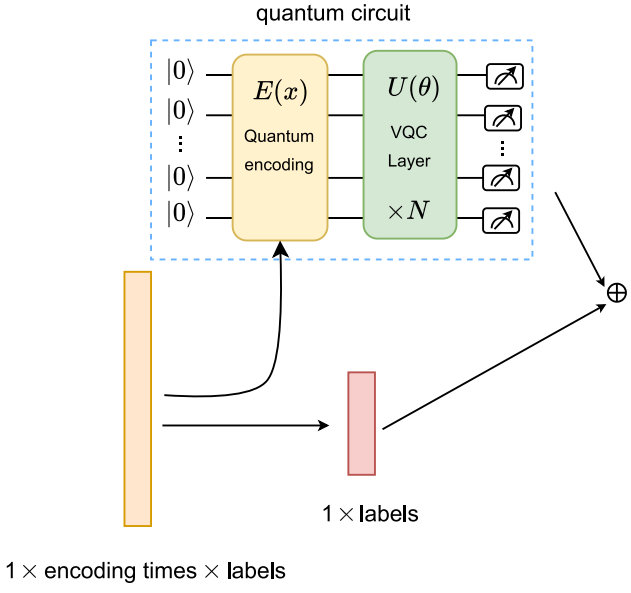


Fig. 6. The structure of quantum–classical residual fusion layer in QMLSC.

5. Experimental results

5.1. Datasets

MELD [49] and IEMOCAP [50] are two specialized datasets developed for multimodal sentiment classification. MELD focusing on classifying dialogue emotions in complex social interactions encountered in real-world scenarios. MELD encompasses data from three

Table 1
The hyperparameters setting in experiments.

Learning rate	Batch size	Encoding	Measurement
3×10^{-5}	4	R_z	Pauli-Z basis

Table 2
Performance comparison of QMLSC with other sentiment classification models on the MELD dataset.

Model	*Accuracy*	$P_{weighted}$	$R_{weighed}$	$F1_{weighted}$	*Params*
Pre+AngEnc	0.6483	0.6214	0.6483	0.6222	108 297
Pre+AmpEnc	0.6529	0.6269	0.6529	0.6170	108 297
Pre+FC	0.6441	0.6216	0.6441	0.6290	308 807
QMNN [51]	0.6081	0.5506	0.6081	0.5800	**76 650**
QMLSC	**0.6586**	**0.6426**	**0.6586**	**0.6365**	108 297

modalities: audio, video, and text, offering a rich variety of data types for researchers to explore different aspects of emotional expression. Each instance is annotated with one of seven primary emotional categories: anger, joy, sadness, surprise, disgust, fear, and neutral. This detailed sentiment classification facilitates a deeper understanding of the intricacies of human emotions. The dataset comprises approximately 13,000 dialogue segments, providing a substantial resource for training large-scale sentiment classification models. IEMOCAP is a specialized dataset focused on analyzing emotions in dyadic interactions. The emotion labels for IEMOCAP are happy, sad, neutral, angry, excited, frustrated. It comprises approximately 12 h of high-quality recordings from both impromptu conversations and scripted performances, with each segment meticulously annotated to ensure accuracy and reliability. IEMOCAP places particular emphasis on the dynamics and continuity of emotional expression, making it an invaluable resource for research in affective computing and human–computer interaction.

5.2. Experimental setting

This study utilized the PyTorch and Pennylane frameworks for model construction and training, executed on an NVIDIA RTX 4090 GPU (24 GB). Audio was extracted from the original dataset's audio and video files using FFmpeg. A quantum circuit was constructed and trained with Pennylane, initializing the state to $|0\rangle$. The hyperparameters used in this study are detailed in Table 1. Due to quantum resource limitations and the effects of sparse plateau phenomena and representation capacity on model performance, this study implemented the model with seven qubits. Furthermore, the learning rate was set to 3×10^{-5}, with a batch size of 4 and an epoch count of 6. The Pauli-Z measurement operator was implemented as the measurement mechanism for the quantum circuit.

5.3. Evaluation metrics

The evaluation metrics used in this study include accuracy, weighted precision, weighted recall, and weighted $F1$ score. Accuracy measures the overall correctness of the model's predictions. Weighted precision evaluates the model's ability to correctly identify positive instances, while weighted recall assesses its ability to capture all relevant positive cases. Finally, the weighted $F1$ score provides a comprehensive evaluation of the model's performance by calculating the weighted average of $F1$ scores across all classes, making it particularly effective for handling class imbalance.

Let n denote the number of categories, with sample counts in each category represented as $N_1, N_2, \ldots, N_n$, and the total sample size is N. The terms used in these metrics are defined as follows:

- TP_i: The number of true positives for category i, i.e., the instances correctly predicted as belonging to category i.
- FP_i: The number of false positives for category i, i.e., the instances incorrectly predicted as belonging to category i.
- TN_i: The number of true negatives for category i, i.e., the instances correctly predicted as not belonging to category i.
- FN_i: The number of false negatives for category i, i.e., the instances incorrectly predicted as not belonging to category i.

Table 3
Performance comparison of QMLSC with other sentiment classification models on the IEMOCAP dataset.

Model	*Accuracy*	$P_{weighted}$	$R_{weighed}$	$F1_{weighted}$	*Params*
Pre+AngEnc	0.6436	0.6585	0.6436	0.6426	92 766
Pre+AmpEnc	0.6484	0.6463	0.6484	0.6460	92 766
Pre+FC	0.6389	0.6376	0.6389	0.6342	308 606
QMNN [51]	0.6084	0.6056	0.6084	0.5988	**52 325**
QMLSC	**0.6531**	**0.6632**	**0.6531**	**0.6490**	92 766

Accuracy is the proportion of correct predictions made by the model out of the total number of predictions. It is computed as:

$$Accuracy = \frac{\sum_{i=1}^{n} TP_i}{N}. \tag{11}$$

Precision for each category i is the proportion of true positives among all instances predicted as belonging to that category. The weighted precision accounts for class imbalance by computing the weighted average of precision across all categories, where the weight is the number of samples in each category N_i. It is calculated as:

$$P_{weighted} = \frac{\sum_{i=1}^{n} P_i \cdot N_i}{N}, \tag{12}$$

where $P_i = \frac{TP_i}{TP_i + FP_i}$ is the precision for category i.

Recall for each category i is the proportion of true positives among all instances that actually belong to that category. Similar to precision, weighted recall is the weighted average of recall values across all categories, weighted by the number of instances in each category. It is defined as:

$$R_{weighted} = \frac{\sum_{i=1}^{n} R_i \cdot N_i}{N}, \tag{13}$$

where $R_i = \frac{TP_i}{TP_i + FN_i}$ is the recall for category i.

The $F1$ score for each category i is the harmonic mean of precision and recall for that category. The weighted $F1$ score takes into account class imbalance by computing the weighted average of $F1$ scores across all categories. It is given by:

$$F1_{weighted} = \frac{\sum_{i=1}^{n} F1_i \cdot N_i}{N}, \tag{14}$$

where $F1_i = 2 \cdot \frac{P_i \cdot R_i}{P_i + R_i}$ is the $F1$ score for category i.

5.4. Model performance

The comparisons between QMLSC and other baselines are summarized in Tables 2 and 3, where *Pre* denotes the data pre-processing component, *AngEnc* (i.e., angle encoding) and *AmpEnc* (i.e., amplitude encoding) represent the substitution of the encoding layer within the quantum circuit layer of the fusion layer module, *Params* denote the number of model parameters following data preprocessing. Additionally, QMNN serves as a contrastive model [51] that also performs sentiment classification by integrating both speech and text data.

As shown in Tables 2 and 3, QMLSC achieves classification accuracies of 65.86% on the MELD dataset and 65.31% on the IEMOCAP dataset, consistently surpassing the four baseline models. Moreover, the classical networks often require multiple rounds of dimensionality reduction to optimize performance, however, these direct approaches may yield suboptimal outcomes. In contrast, our proposed quantum coding necessitates only a single reduction to achieve the desired feature count, thereby minimizing the number of parameters. Within the

Table 4
Experimental result of the ablation study.

Dataset	Model		*Accuracy*	$P_{weighted}$	$R_{weighed}$	$F1_{weighted}$
MELD	QMLSC		**0.6586**	**0.6426**	**0.6586**	**0.6365**
	w/o QCRF[a]	rmv QP[c]	0.6326	0.5998	0.6326	0.6008
		rmv CP[d]	0.6207	0.5327	0.6207	0.5575
	w/o QCSO[b]	tmpl 1[e]	0.6433	0.6215	0.6433	0.6177
		tmpl 2	0.6464	0.6161	0.6464	0.6212
		tmpl 3	0.6506	0.6311	0.6506	0.6317
		tmpl 4	0.6448	0.6160	0.6448	0.6212
		tmpl 5	0.6464	0.6202	0.6464	0.6194
		tmpl 6	0.6490	0.6192	0.6490	0.6245
IEMOCAP	QMLSC		**0.6531**	**0.6632**	**0.6532**	**0.6490**
	w/o QCRF	rmv QP	0.6179	0.6158	0.6179	0.6096
		rmv CP	0.6091	0.6123	0.6091	0.5930
	w/o QCSO	tmpl 1	0.6301	0.6319	0.6301	0.6268
		tmpl 2	0.6436	0.6401	0.6436	0.6391
		tmpl 3	0.6375	0.6432	0.6375	0.6329
		tmpl 4	0.6382	0.6437	0.6382	0.6348
		tmpl 5	0.6402	0.6411	0.6402	0.6371
		tmpl 6	0.6430	0.6425	0.6430	0.6389

[a] QMLSC without the structure of quantum–classical residual fusion layer.
[b] QMLSC without performing the quantum circuit structure optimization.
[c] QMLSC that removing the quantum part in quantum–classical residual fusion layer.
[d] QMLSC that removing the classical part in quantum–classical residual fusion layer.
[e] QMLSC with the VQC layer replaced by fixed circuit template 1.

same preprocessing framework, quantum fusion layers demonstrate significant advantages in parameter efficiency compared to conventional fully connected classical layers, requiring approximately one-third of the parameters (64.9% reduction in total parameter count on MELD dataset and 69.9% reduction on IEMOCAP dataset). This mechanism underscores the benefits of quantum feature processing.

The quantum–classical residual fusion layer and quantum circuit structure optimization are critical modules of QMLSC. To rigorously evaluate their significance, comprehensive ablation experiments were conducted, providing a systematic assessment of their contributions. Table 4 details the changes in network performance upon removing different components and compares the effects of various quantum circuit templates illustrated in Fig. 7. The experimental results demonstrate that without quantum–classical residual connections, quantum circuits are more susceptible to the "barren plateau" phenomenon [52,53], characterized by vanishing gradients, which complicates the training process and necessitates additional training epochs to achieve satisfactory performance. However, with the introduction of the quantum residual layer, both the training duration and model convergence are significantly improved, leading to enhanced overall efficiency.

Obviously, removing either the classical or quantum components significantly degrades model performance, confirming the substantial contributions of the quantum–classical residual fusion layer to model efficiency. It also highlights the capability of VQCs to map classical features into a higher-dimensional and more expressive feature space. Moreover, while replacing the VQC layer with fixed circuit templates has a relatively minor impact on accuracy, the experimental results suggest that optimizing quantum circuits holds some importance for improving model performance.

5.5. Robustness analysis against quantum noise

In the era of NISQ computing, understanding and managing quantum noise is pivotal for maintaining the performance of quantum machine learning (QML) models. Quantum noise arises from interactions between quantum systems and their environments, operational inaccuracies, and random fluctuations of quantum states, leading to the degradation of quantum states. This inherent physical phenomenon can distort the decision boundaries in quantum classification tasks and diminish accuracy. Additionally, quantum noise has been shown to reduce model performance by inducing the "barren plateau" phenomenon [54,55], where gradients approach zero across most regions of the parameter space. This complicates the optimization process, hindering algorithms from identifying effective update directions. As a result, assessing and mitigating the impact of quantum noise is critical for advancing QML in practical applications.

One approach to understanding quantum noise involves mathematical modeling at the single-qubit level. Specifically, quantum noise associated with a single qubit can be effectively described using Kraus operators. In quantum computing, several common types of quantum noise are frequently encountered, including phase flip, bit flip, amplitude damping, and depolarizing noise. Each of these noise models captures distinct physical processes that can disrupt quantum states and impact computation fidelity.

Phase flip noise alters the relative phase of the quantum state while leaving the basis states unchanged. For instance, $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \rightarrow |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. The Kraus operators of phase flip (PF) noise are

$$K_0 = \sqrt{1-p}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, \quad K_1 = \sqrt{p}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}. \tag{15}$$

Bit flip noise is a random flip between the states $|0\rangle$ and $|1\rangle$, that is, $|0\rangle \rightarrow |1\rangle$ or $|1\rangle \rightarrow |0\rangle$. The Kraus operators of bit flip (BF) noise are

$$K_0 = \sqrt{1-p}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, \quad K_1 = \sqrt{p}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}. \tag{16}$$

Amplitude damping channel simulates the attenuation process of energy from the excited state $|1\rangle$ to the ground state $|0\rangle$, similar to the process of spontaneous emission in atoms. The Kraus operators of amplitude damping (AD) noise are

$$K_0 = \begin{bmatrix}1 & 0\\ 0 & \sqrt{1-p}\end{bmatrix}, \quad K_1 = \begin{bmatrix}0 & \sqrt{p}\\ 0 & 0\end{bmatrix}. \tag{17}$$

Depolarizing noise characterizes the process by which a quantum system evolves into a completely mixed state. With probability p, the state of a qubit is randomly mapped to one of the outcomes resulting from the action of the Pauli operators I, X, Y, or Z, while with probability $1-p$, it remains unchanged. Here, I denotes the identity operator,

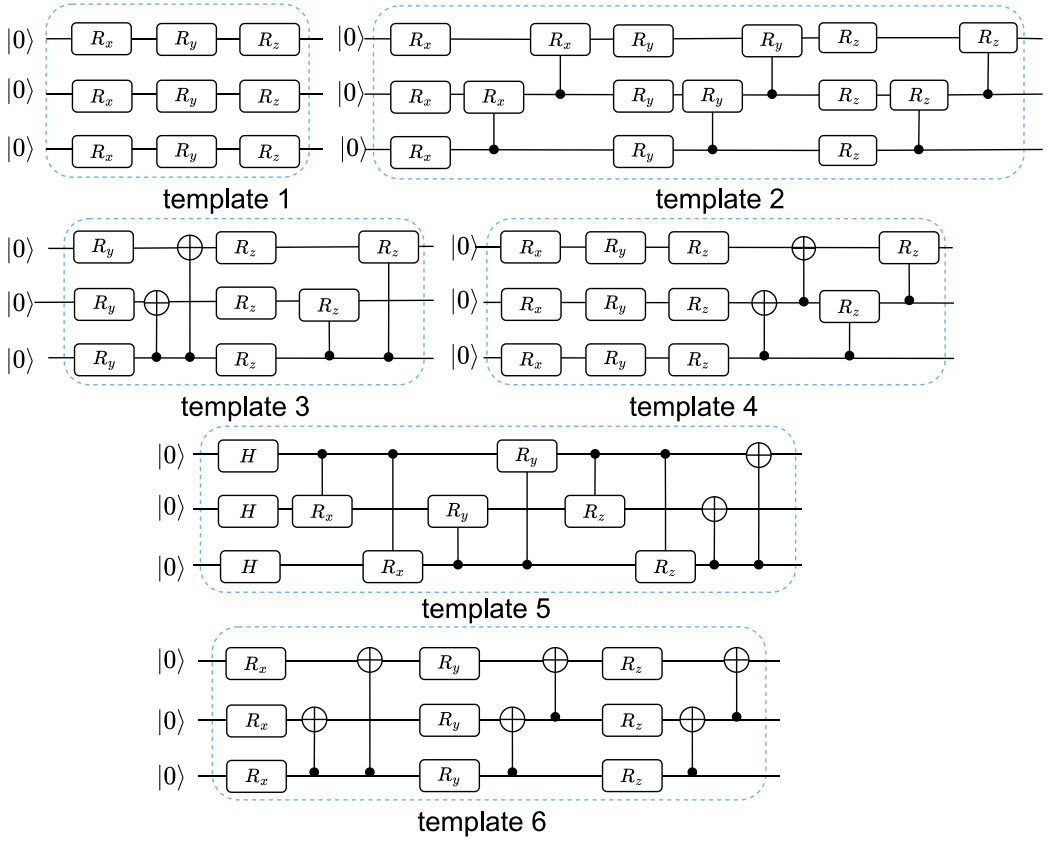


Fig. 7. The template set consists of circuits inspired by or derived from previous research, with dashed boxes denoting individual circuit layers that can be repeated.

and I, X, Y, and Z represent the Pauli-X, Pauli-Y, and Pauli-Z matrices, respectively. The Kraus operators of depolarization noise (DN) are

$$K_0 = \sqrt{1-p}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}, \quad K_1 = \sqrt{\frac{p}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix},$$
$$K_2 = \sqrt{\frac{p}{3}}\begin{bmatrix}0 & -i\\ i & 0\end{bmatrix}, \quad K_3 = \sqrt{\frac{p}{3}}\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}, \tag{18}$$

where $p \in [0, 1]$ denotes the probability of noise occurring on a single qubit.

Building on the discussion of various types of quantum noise and their mathematical representations, it is essential to describe how these noise effects influence the evolution of quantum states. This evolution can be captured using the formalism of the density matrix, which provides a comprehensive framework for modeling both pure and mixed states under noise. Specifically, the impact of quantum noise on a quantum state can be expressed through a quantum operation, defined as:

$$\mathcal{E}(|\psi\rangle\langle\psi|) = \sum_i K_i|\psi\rangle\langle\psi|K_i^\dagger \tag{19}$$

where K_i are Kraus operators that characterize the noise channels. The Hermitian conjugate $K_i^\dagger$ ensures the completeness relation $\sum_i K_i^\dagger K_i = I$, preserving the trace of the density matrix.

Table 5 illustrates the performance of the model on two datasets under the influence of four types of noise, both individually and in combination (i.e., ALL). The results reveal that training QMLSC with noise intensity p of 0.10 and 0.01 yields accuracy levels comparable to those in noise-free conditions, highlighting the remarkable resistance to noise interference.

To thoroughly investigate the potential reasons behind the robustness of QMLSC, it is suggested that its robustness primarily arises from the quantum–classical hybrid residual structure. This structure introduces skip connections that enable direct information flow between layers, significantly enhancing the model's expressive capability and overall stability. Additionally, the module's unique design allows the gradient descent algorithm to dynamically adjust the weight distribution between quantum and classical features during training, further optimizing its performance. To validate this hypothesis, we conducted detailed experiments evaluating the performance of templates 2 and 4 under noisy conditions. The results demonstrate that both templates consistently achieve high classification accuracy despite quantum noise interference. Performance data across various noise levels, as shown in Tables 6, further corroborate the critical role of the quantum–classical hybrid residual structure in enhancing the model's robustness.

6. Conclusion

This study proposes QMLSC, a multi-modal fusion multi-task learning algorithm that integrates classical and quantum networks within a residual architecture. The model employs self-attention and cross-attention mechanisms to achieve deep integration of multi-modal features. To address quantum resource limitations, our model utilizes a batch-coding feature strategy, sequentially inputting classical data into the quantum circuit across multiple stages, optimizing resource allocation and utilization. The algorithm innovatively fuses quantum and classical features within the residual framework and uses a random search approach to optimize the quantum circuit structure. This technique significantly enhances flexibility, adaptability, task-specific optimization, learning efficiency, and generalization capability. Experimental results indicate that QMLSC outperforms conventional fully connected layers in accuracy while reducing parameter count by over 64%. This methodology improves the model's capacity to handle complex data, mitigates the impact of interference from quantum noise,

Table 5
Performance of the QMLSC model under different quantum noise types and intensities p.

DataSet	Model	P	Accuracy	$P_{weighted}$	$R_{weighted}$	$F1_{weighted}$
MELD	PF	0.01	0.6513	0.6275	0.6513	0.6332
		0.10	0.6487	0.6228	0.6487	0.6284
	BF	0.01	0.6517	0.6262	0.6517	0.6318
		0.10	0.6544	0.6310	0.6544	0.6360
	AD	0.01	0.6483	0.6211	0.6483	0.6274
		0.10	0.6517	0.6273	0.6517	0.6305
	DN	0.01	0.6529	0.6284	0.6529	0.6348
		0.10	0.6513	0.6288	0.6513	0.6211
	ALL	0.01	0.6498	0.6253	0.6498	0.6308
		0.10	0.6513	0.6268	0.6513	0.6326
IEMOCAP	PF	0.01	0.6301	0.6352	0.6301	0.6218
		0.10	0.6328	0.6374	0.6328	0.6237
	BF	0.01	0.6545	0.6641	0.6545	0.6504
		0.10	0.6545	0.6643	0.6545	0.6505
	AD	0.01	0.6355	0.6392	0.6355	0.6329
		0.10	0.6396	0.6436	0.6396	0.6389
	DN	0.01	0.6362	0.6430	0.6362	0.6344
		0.10	0.6335	0.6398	0.6335	0.6317
	ALL	0.01	0.6558	0.6662	0.6558	0.6520
		0.10	0.6301	0.6339	0.6301	0.6215

Table 6
Accuracy of the QMLSC model with VQC layer replaced by fixed circuit templates 2 or 4 under different quantum noise types and intensities p.

Model	Noise type	p	*Accuracy*	
			MELD	IEMOCAP
QMLSC with the template 2	PF	0.01	0.6418	0.6301
		0.10	0.6421	0.6362
	BF	0.01	0.6452	0.6545
		0.10	0.6467	0.6267
	AD	0.01	0.6418	0.6355
		0.10	0.6402	0.6382
	DN	0.01	0.6464	0.6362
		0.10	0.6464	0.6362
	ALL	0.01	0.6429	0.6558
		0.10	0.6418	0.6369
QMLSC with the template 4	PF	0.01	0.6433	0.6369
		0.10	0.6421	0.6226
	BF	0.01	0.6437	0.6240
		0.10	0.6437	0.6240
	AD	0.01	0.6429	0.6382
		0.10	0.6398	0.6220
	DN	0.01	0.6444	0.6375
		0.10	0.6425	0.6274
	ALL	0.01	0.6387	0.6382
		0.10	0.6418	0.6389

retaining the unique benefits of quantum computing while leveraging the stability and maturity of classical computing.

CRediT authorship contribution statement

YaoChong Li: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Yi Qu:** Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Ri-Gui Zhou:** Supervision, Resources, Project administration, Investigation, Funding acquisition. **Jing Zhang:** Visualization, Validation, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] R. Das, T.D. Singh, Multimodal sentiment analysis: a survey of methods, trends, and challenges, ACM Comput. Surv. 55 (13s) (2023) 1–38.
[2] Y. Huang, C. Du, Z. Xue, X. Chen, H. Zhao, L. Huang, What makes multimodal learning better than single (provably), in: Proceedings of the 35th International Conference on Neural Information Processing Systems, vol. 34, 2021, pp. 10944–10956.
[3] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, S. Lloyd, Quantum machine learning, Nature 549 (7671) (2017) 195–202.
[4] V. Havlíček, A.D. Córcoles, K. Temme, A.W. Harrow, A. Kandala, J.M. Chow, J.M. Gambetta, Supervised learning with quantum-enhanced feature spaces, Nature 567 (7747) (2019) 209–212.
[5] V. Dunjko, J.M. Taylor, H.J. Briegel, Quantum-enhanced machine learning, Phys. Rev. Lett. 117 (2016) 130501.
[6] P. Tiwari, L. Zhang, Z. Qu, G. Muhammad, Quantum fuzzy neural network for multimodal sentiment and sarcasm detection, Inf. Fusion 103 (2024) 102085.
[7] Z. Qu, Y. Meng, G. Muhammad, P. Tiwari, Qmfnd: A quantum multimodal fusion-based fake news detection model for social media, Inf. Fusion 104 (2024) 102172.
[8] Z. Qu, Y. Li, P. Tiwari, Qnmf: A quantum neural network based multimodal fusion system for intelligent diagnosis, Inf. Fusion 100 (2023) 101913.
[9] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, J.R. McClean, Power of data in quantum machine learning, Nat. Commun. 12 (1) (2021) 2631.
[10] I. Cong, S. Choi, M.D. Lukin, Quantum convolutional neural networks, Nat. Phys. 15 (12) (2019) 1273–1278.
[11] J. Zhao, W. Qiao, P. Zhang, H. Gao, Quantum implicit neural representations, in: Proceedings of the 41st International Conference on Machine Learning, ICML'24, 2024, pp. 60940–60956.
[12] G. Torlai, R.G. Melko, Machine-learning quantum states in the nisq era, Annu. Rev. Condens. Matter Phys. 11 (2020) 325–344.
[13] H. Liao, D.S. Wang, I. Sitdikov, C. Salcedo, A. Seif, Z.K. Minev, Machine learning for practical quantum error mitigation, Nat. Mach. Intell. 6 (12) (2024) 1478–1486.

[14] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.

[15] V. Havlíček, A.D. Córcoles, K. Temme, A.W. Harrow, A. Kandala, J.M. Chow, J.M. Gambetta, Supervised learning with quantum-enhanced feature spaces, Nature 567 (7747) (2019) 209–212.

[16] C. Ciliberto, M. Herbster, A.D. Ialongo, M. Pontil, A. Rocchetto, S. Severini, L. Wossnig, Quantum machine learning: a classical perspective, Proc. R. Soc. A: Math. Phys. Eng. Sci. 474 (2018) 20170551.

[17] K. Beer, D. Bondarenko, T. Farrelly, T.J. Osborne, R. Salzmann, D. Scheiermann, R. Wolf, Training deep quantum neural networks, Nat. Commun. 11 (2020).

[18] D. Tang, B. Qin, X. Feng, T. Liu, Effective lstms for target-dependent sentiment classification, in: 26th International Conference on Computational Linguistics, 2016, pp. 3298–3307.

[19] Y. Wang, M. Huang, X. Zhu, L. Zhao, Attention-based LSTM for aspect-level sentiment classification, in: Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing, 2016, pp. 606–615.

[20] Z. Zhou, F. Liu, Filter gate network based on multi-head attention for aspect-level sentiment classification, Neurocomputing 441 (2021) 214–225.

[21] C. Zhang, Q. Li, D. Song, Aspect-based sentiment classification with aspect-specific graph convolutional networks, 2019, arXiv:arXiv:1909.03477.

[22] T.N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, 2016, arXiv:arXiv:1609.02907.

[23] Y. Shan, C. Che, X. Wei, X. Wang, Y. Zhu, B. Jin, Bi-graph attention network for aspect category sentiment classification, Knowl.-Based Syst. 258 (2022) 109972.

[24] D. Ma, S. Li, X. Zhang, H. Wang, Interactive attention networks for aspect-level sentiment classification, in: Proceedings of the 26th International Joint Conference on Artificial Intelligence, 2017, pp. 4068–4074.

[25] T. Gu, H. Zhao, Z. He, M. Li, D. Ying, Integrating external knowledge into aspect-based sentiment analysis using graph neural network, Knowl.-Based Syst. 259 (2023) 110025.

[26] G. Lu, J. Li, J. Wei, Aspect sentiment analysis with heterogeneous graph neural networks, Inf. Process. Manage. 59 (4) (2022) 102953.

[27] N. Majumder, S. Poria, A.F. Gelbukh, M.S. Akhtar, E. Cambria, A. Ekbal, IARM: inter-aspect relation modeling with memory networks in aspect-based sentiment analysis, in: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 2018, pp. 3402–3411.

[28] D. Hazarika, S. Poria, A. Zadeh, E. Cambria, L. Morency, R. Zimmermann, Conversational memory network for emotion recognition in dyadic dialogue videos, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 2018, pp. 2122–2132.

[29] S.C. Kak, Quantum neural computing, Adv. Imaging Electron Phys. 94 (1995) 259–313.

[30] D. Ventura, T.R. Martinez, An artificial neuron with quantum mechanical properties, in: Artificial Neural Nets and Genetic Algorithms, 1998, pp. 482–485.

[31] N. Matsui, M. Takai, H. Nishimura, A network model based on qubitlike neuron corresponding to quantum circuit, Electron. Commun. Jpn. (Part III: Fundam. Electron. Science) 83 (10) (2000) 67–73.

[32] R. Zhou, L. Qin, N. Jiang, Quantum perceptron network, in: Artificial Neural Networks – ICANN, 2006, pp. 651–657.

[33] E. Farhi, H. Neven, Classification with quantum neural networks on near term processors, 2018, arXiv:arXiv:1802.06002.

[34] I. Cong, S. Choi, M.D. Lukin, Quantum convolutional neural networks, Nat. Phys. 15 (12) (2019) 1273–1278.

[35] Y. Du, M.-H. Hsieh, T. Liu, D. Tao, N. Liu, Quantum noise protects quantum classifiers against adversaries, Phys. Rev. Res. 3 (2) (2021) 023153.

[36] Z. Qu, X. Liu, M. Zheng, Temporal-spatial quantum graph convolutional neural network based on Schrödinger approach for traffic congestion prediction, IEEE Trans. Intell. Transp. Syst. 24 (8) (2022) 8677–8686.

[37] Y. Zhang, D. Song, X. Li, P. Zhang, P. Wang, L. Rong, G. Yu, B. Wang, A quantum-like multimodal network framework for modeling interaction dynamics in multiparty conversational sentiment analysis, Inf. Fusion 62 (2020) 14–31.

[38] K. Meichanetzidis, A. Toumi, G. de Felice, B. Coecke, Grammar-aware sentence classification on quantum computers, Quantum Mach. Intell. 5 (2023) 10.

[39] R. Lorenz, A. Pearson, K. Meichanetzidis, D. Kartsaklis, B. Coecke, Qnlp in practice: Running compositional models of meaning on a quantum computer, J. Artificial Intelligence Res. 76 (2023) 1305–1342.

[40] D. Gkoumas, Q. Li, S. Dehdashti, M. Melucci, Y. Yu, D. Song, Quantum cognitively motivated decision fusion for video sentiment analysis, in: Thirty-Fifth AAAI Conference on Artificial Intelligence, 2021, pp. 827–835.

[41] D. Gkoumas, Q. Li, Y. Yu, D. Song, An entanglement-driven fusion neural network for video sentiment analysis, in: Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence, 2021, pp. 1736–1742.

[42] Y. Liu, Y. Zhang, Q. Li, B. Wang, D. Song, What does your smile mean? Jointly detecting multi-modal sarcasm and sentiment using quantum probability, in: Findings of the Association for Computational Linguistics: EMNLP 2021, 2021, pp. 871–880.

[43] Y. Zhang, D. Song, P. Zhang, X. Li, P. Wang, A quantum-inspired sentiment representation model for twitter sentiment analysis, Appl. Intell. 49 (2019) 3093–3108.

[44] J. Devlin, M. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, 2018, arXiv:arXiv:1810.04805.

[45] W.-N. Hsu, B. Bolte, Y.-H.H. Tsai, K. Lakhotia, R. Salakhutdinov, A. Mohamed, Hubert: Self-supervised speech representation learning by masked prediction of hidden units, IEEE/ACM Trans. Audio Speech Lang. Process. 29 (2021) 3451–3460.

[46] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, J.I. Latorre, Data re-uploading for a universal quantum classifier, Quantum 4 (2020) 226.

[47] P. Easom-McCaldin, A. Bouridane, A. Belatreche, R. Jiang, S. Al-Maadeed, Efficient quantum image classification using single qubit encoding, IEEE Trans. Neural Netw. Learn. Syst. 35 (2024) 1472–1486.

[48] M. Schuld, A. Bocharov, K.M. Svore, N. Wiebe, Circuit-centric quantum classifiers, Phys. Rev. A 101 (2020) 032308.

[49] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, R. Mihalcea, Meld: A multimodal multi-party dataset for emotion recognition in conversations, 2018, arXiv:arXiv:1810.02508.

[50] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J.N. Chang, S. Lee, S.S. Narayanan, IEMOCAP: interactive emotional dyadic motion capture database, Lang. Resour. Eval. 42 (4) (2008) 335–359.

[51] Q. Li, D. Gkoumas, A. Sordoni, J. Nie, M. Melucci, Quantum-inspired neural network for conversational emotion recognition, in: Thirty-Fifth AAAI Conference on Artificial Intelligence, 2021, pp. 13270–13278.

[52] J.R. McClean, S. Boixo, V.N. Smelyanskiy, R. Babbush, H. Neven, Barren plateaus in quantum neural network training landscapes, Nat. Commun. 9 (1) (2018) 4812.

[53] A. Pesah, M. Cerezo, S. Wang, T. Volkoff, A.T. Sornborger, P.J. Coles, Absence of barren plateaus in quantum convolutional neural networks, Phys. Rev. X 11 (2021) 041011.

[54] S. Wang, E. Fontana, M. Cerezo, K. Sharma, A. Sone, L. Cincio, P.J. Coles, Noise-induced barren plateaus in variational quantum algorithms, Nat. Commun. 12 (1) (2021) 6961.

[55] M. Cerezo, G. Verdon, H.-Y. Huang, L. Cincio, P.J. Coles, Challenges and opportunities in quantum machine learning, Nat. Comput. Sci. 2 (9) (2022) 567–576.